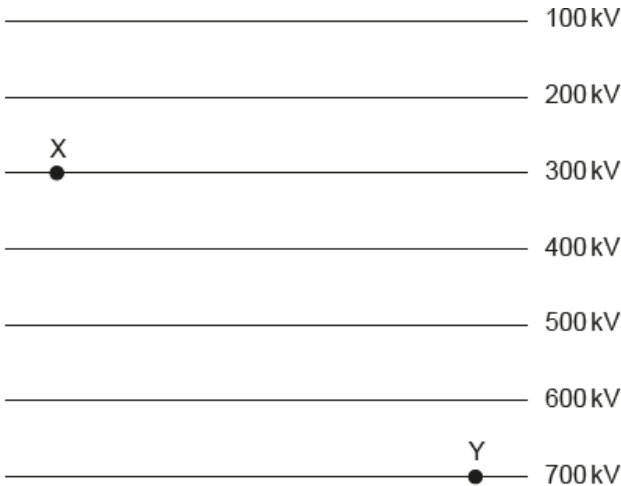


Topic D1+D2 HL Paper 1A

Study Guide [28 marks] -- markscheme

1. A particle with charge $-2.5 \times 10^{-6} \text{ C}$ moves from point X to point Y due to [1 mark] a uniform electrostatic field. The diagram shows some equipotential lines of the field.



What is correct about the motion of the particle from X to Y and the magnitude of the work done by the field on the particle?

	Motion of the particle from X to Y	Magnitude of the work done by the field on the particle
A.	uniform linear	0 J
B.	uniform linear	1 J
C.	uniformly accelerated	0 J
D.	uniformly accelerated	1 J

Markscheme

D

2. Which is a correct unit for gravitational potential? [1 mark]
- A. $\text{m}^2 \text{s}^{-2}$
 - B. J kg
 - C. m s^{-2}
 - D. $\text{N m}^{-1} \text{kg}^{-1}$

Markscheme

A

3. A planet has radius R . The escape speed from the surface of the planet is v . At what distance from the surface of the planet is the orbital speed $0.5v$? [1 mark]
- A. $0.5R$
 - B. R
 - C. $2R$
 - D. $4R$

Markscheme

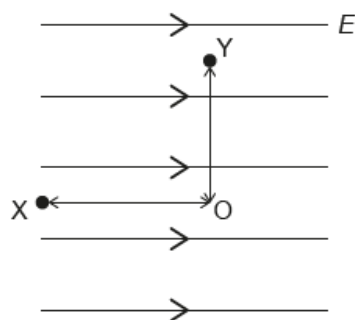
B

4. An object of mass m released from rest near the surface of a planet has an initial acceleration z . What is the gravitational field strength near the surface of the planet? [1 mark]
- A. z
 - B. $\frac{z}{m}$
 - C. mz
 - D. $\frac{m}{z}$

Markscheme

A

5. The points X and Y are in a uniform electric field of strength E . The distance OX is x and the distance OY is y . [1 mark]



What is the magnitude of the change in electric potential between X and Y?

- A. Ex
- B. Ey
- C. $E(x + y)$
- D. $E\sqrt{x^2 + y^2}$

Markscheme

A

6. A satellite orbits planet X with a speed v_x at a distance r from the centre [1 mark] of planet X. Another satellite orbits planet Y at a speed of v_y at a distance r from the centre of planet Y. The mass of planet X is M and the mass of planet Y is $4M$. What is the ratio of $\frac{v_x}{v_y}$?

- A. 0.25
- B. 0.5
- C. 2.0
- D. 4.0

Markscheme

B

7. The escape velocity for an object at the surface of the Earth is v_{esc} . The diameter of the Moon is 4 times smaller than that of the Earth and the mass of the Moon is 81 times smaller than that of the Earth. What is the escape velocity of the object on the Moon? [1 mark]

- A. $\frac{2}{81} v_{\text{esc}}$
B. $\frac{4}{81} v_{\text{esc}}$
C. $\frac{2}{9} v_{\text{esc}}$
D. $\frac{4}{9} v_{\text{esc}}$

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Markscheme

C

8. A satellite in a circular orbit around the Earth needs to reduce its orbital radius. [1 mark]

What is the work done by the satellite rocket engine and the change in kinetic energy resulting from this shift in orbital height?

	Work done by the satellite rocket engine	Kinetic energy
A.	positive	increase
B.	positive	decrease
C.	negative	increase
D.	negative	decrease

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Markscheme

C

9. The escape speed from a planet of radius R is v_{esc} . A satellite orbits the planet at a distance R from the surface of the planet. What is the orbital speed of the satellite? [1 mark]

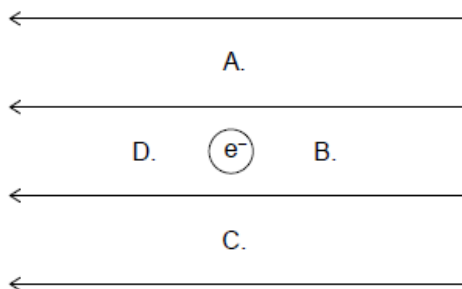
- A. $\frac{1}{2}v_{\text{esc}}$
B. $\frac{\sqrt{2}}{2}v_{\text{esc}}$
C. $\sqrt{2}v_{\text{esc}}$
D. $2v_{\text{esc}}$

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Markscheme

A

10. An electron is fixed in position in a uniform electric field. What is the position for which the electrical potential energy of the electron is greatest? [1 mark]

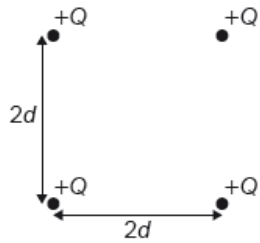


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Markscheme

D

11. Four identical, positive, point charges of magnitude Q are placed at the vertices of a square of side $2d$. What is the electric potential produced at the centre of the square by the four charges? *[1 mark]*

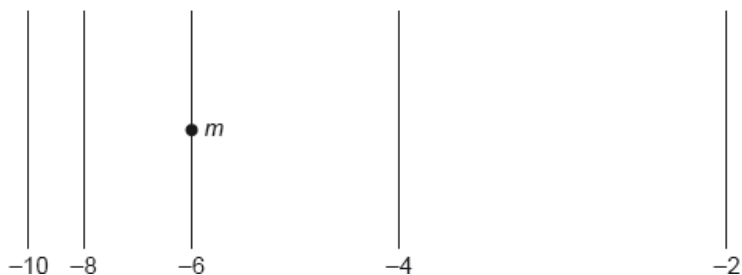


- A. 0
B. $\frac{4kQ}{d}$
C. $\frac{\sqrt{2}kQ}{d}$
D. $\frac{2\sqrt{2}kQ}{d}$

Markscheme

D

12. The diagram shows 5 gravitational equipotential lines. The gravitational potential on each line is indicated. A point mass m is placed on the middle line and is then released. Values given in MJ kg⁻¹. [1 mark]



Which is correct about the direction of motion and the acceleration of the point mass?

	Direction	Acceleration
A.	to the right	decreasing
B.	to the right	increasing
C.	to the left	decreasing
D.	to the left	increasing

Markscheme

D

13. An electron of mass m_e orbits an alpha particle of mass m_α in a circular orbit of radius r . Which expression gives the speed of the electron? [1 mark]

- A. $\sqrt{\frac{2ke^2}{m_e r}}$
 B. $\sqrt{\frac{2ke^2}{m_\alpha r}}$
 C. $\sqrt{\frac{4ke^2}{m_e r}}$
 D. $\sqrt{\frac{4ke^2}{m_\alpha r}}$

Markscheme

A

14. A moon of mass M orbits a planet of mass $100M$. The radius of the planet is R and the distance between the centres of the planet and moon is $22R$. [1 mark]



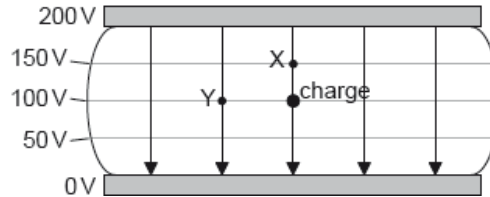
What is the distance from the centre of the planet at which the total gravitational potential has a maximum value?

- A. $2R$
- B. $11R$
- C. $20R$
- D. $2R$ and $20R$

Markscheme

C

15. The diagram shows the electric field and the electric equipotential surfaces between two charged parallel plates. The potential difference between the plates is 200 V. [1 mark]



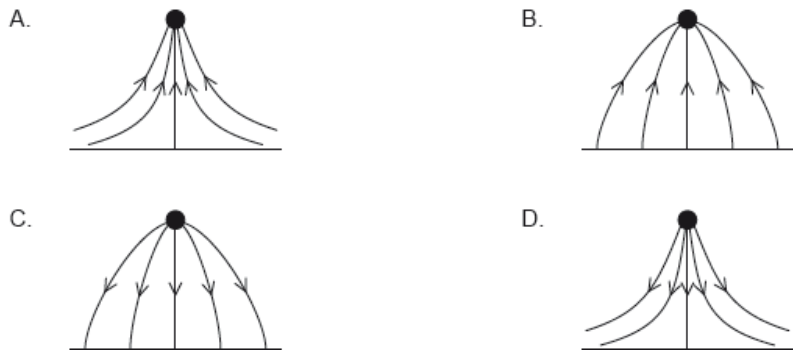
What is the work done, in nJ, by the electric field in moving a negative charge of magnitude 1 nC from the position shown to X and to Y?

- | | To X | To Y |
|----|------|------|
| A. | 50 | 0 |
| B. | -50 | 0 |
| C. | 50 | 100 |
| D. | -50 | -100 |

Markscheme

A

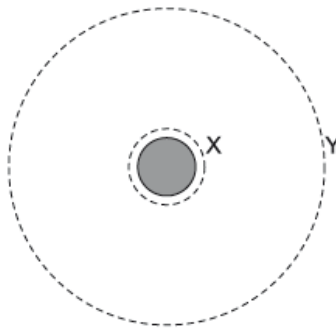
16. A positive point charge is placed above a metal plate at zero electric potential. Which diagram shows the pattern of electric field lines between the charge and the plate? [1 mark]



Markscheme

C

17. A satellite orbiting a planet moves from orbit X to orbit Y. [1 mark]



What is the change in the kinetic energy and the change in the gravitational potential energy as a result?

	Kinetic energy	Gravitational potential energy
A.	increases	increases
B.	increases	decreases
C.	decreases	increases
D.	decreases	decreases

Markscheme

C

18. The mass of the Earth is M_E and the mass of the Moon is M_M . Their respective radii are R_E and R_M .

[1 mark]

Which is the ratio $\frac{\text{escape speed from the Earth}}{\text{escape speed from the Moon}}$?

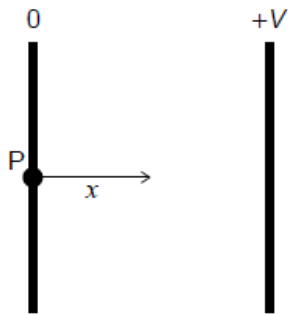
- A. $\sqrt{\frac{M_M R_M}{M_E R_E}}$
 B. $\sqrt{\frac{M_E R_E}{M_M R_M}}$
 C. $\sqrt{\frac{M_E R_M}{M_M R_E}}$
 D. $\sqrt{\frac{M_M R_E}{M_E R_M}}$

Markscheme

C

19. An electric field acts in the space between two charged parallel plates. One plate is at zero potential and the other is at potential $+V$.

[1 mark]



The distance x is measured from point P in the direction perpendicular to the plate.

What is the dependence of the electric field strength E on x and what is the dependence of the electric potential V on x ?

	E	V
A.	proportional to $\frac{1}{x^2}$	constant
B.	constant	proportional to x
C.	proportional to x	proportional to x
D.	proportional to x^2	constant

Markscheme

B

20. A satellite at the surface of the Earth has a weight W and gravitational potential energy E_p . The satellite is then placed in a circular orbit with a radius twice that of the Earth. [1 mark]

What is the weight of the satellite and the gravitational potential energy of the satellite when placed in orbit?

	Weight	Gravitational potential energy
A.	$0.25 W$	$0.25 E_p$
B.	$0.5 W$	$0.25 E_p$
C.	$0.25 W$	$0.5 E_p$
D.	$0.5 W$	$0.5 E_p$

Markscheme

C

21. Two point charges are at rest as shown. [1 mark]

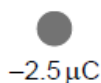
At which position is the electric field strength greatest?

C.

A.



B.

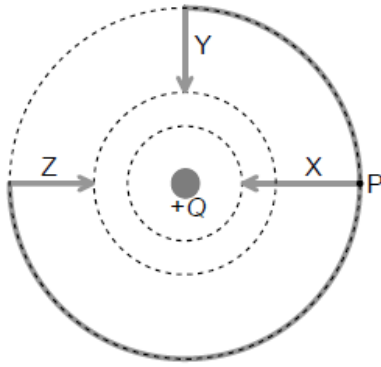


D.

Markscheme

B

22. A positive charge Q is deposited on the surface of a small sphere. The dotted lines represent equipotentials. [1 mark]



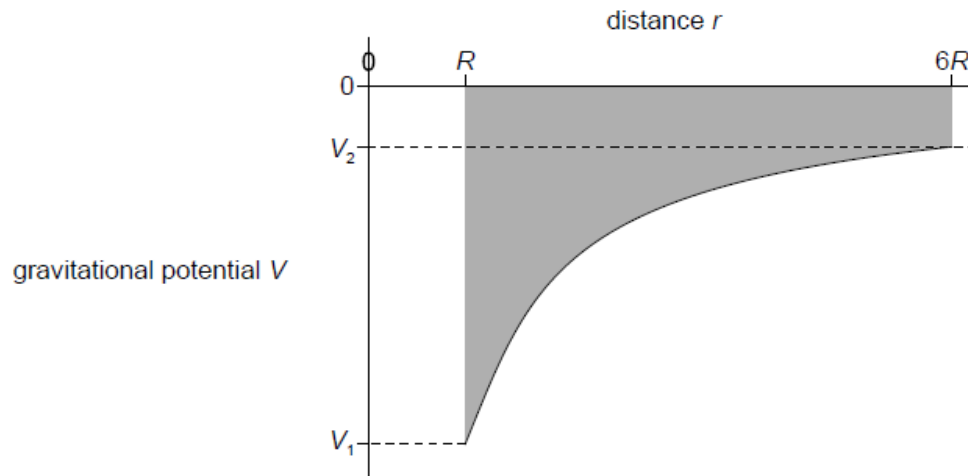
A small positive point charge is moved from point P closer to the sphere along three different paths X, Y and Z. The work done along each path is W_X , W_Y and W_Z . What is a correct comparison of W_X , W_Y and W_Z ?

- A. $W_Z > W_Y > W_X$
- B. $W_X > W_Y = W_Z$
- C. $W_X = W_Y = W_Z$
- D. $W_Z = W_Y > W_X$

Markscheme

B

23. The graph shows the variation of the gravitational potential V with distance r from the centre of a uniform spherical planet. The radius of the planet is R . The shaded area is S . [1 mark]



What is the work done by the gravitational force as a point mass m is moved from the surface of the planet to a distance $6R$ from the centre?

- A. $m (V_2 - V_1)$
- B. $m (V_1 - V_2)$
- C. mS
- D. S

Markscheme

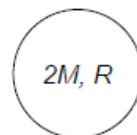
B

24. Four uniform planets have masses and radii as shown. Which planet has the smallest escape speed? [1 mark]

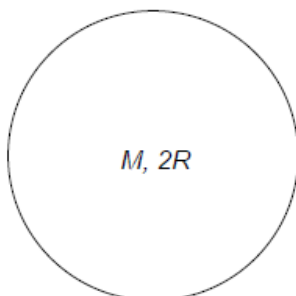
A.



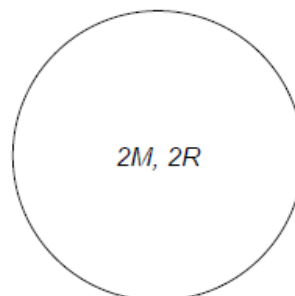
B.



C.



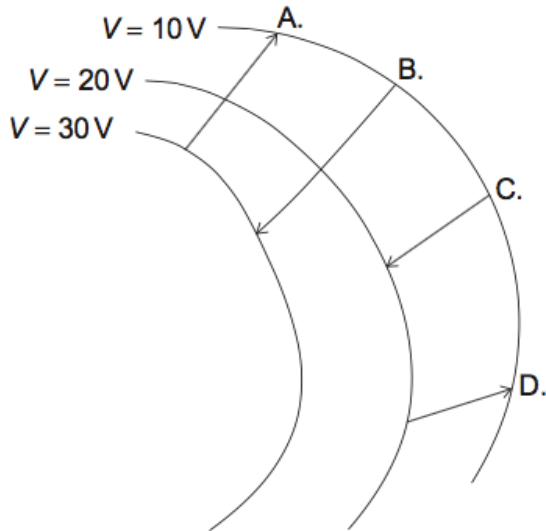
D.



Markscheme

C

25. A negative charge moves in an electric field. Equipotential lines for the field and four possible paths of the charge are shown. Which path corresponds to the largest work done on the charge by the field? [1 mark]



Markscheme

B

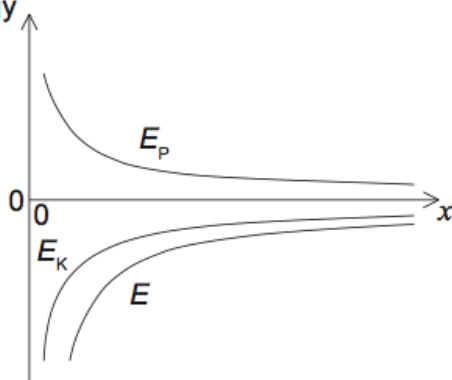
26. In an experiment, oil droplets of mass m and charge q are dropped into the region between two horizontal parallel plates. The electric field E between the plates can be adjusted. Air resistance is negligible. Which is correct when the droplets fall vertically at constant velocity? [1 mark]
- A. $E=0$
 - B. $E < \frac{mg}{q}$
 - C. $E = \frac{mg}{q}$
 - D. $E > \frac{mg}{q}$

Markscheme

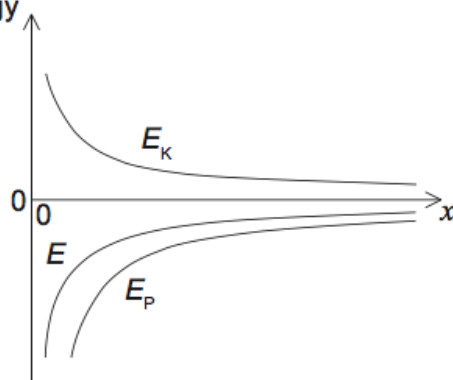
C

27. A satellite orbits a planet. Which graph shows how the kinetic energy E_K , [1 mark] the potential energy E_P and the total energy E of the satellite vary with distance x from the centre of the planet?

A. energy

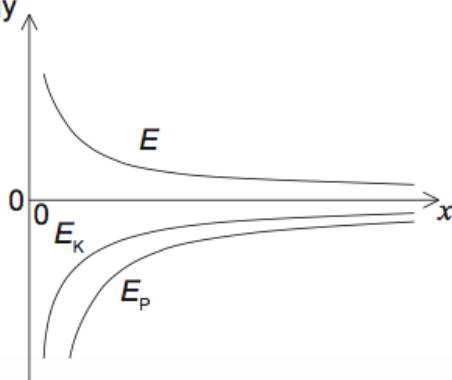


B. energy



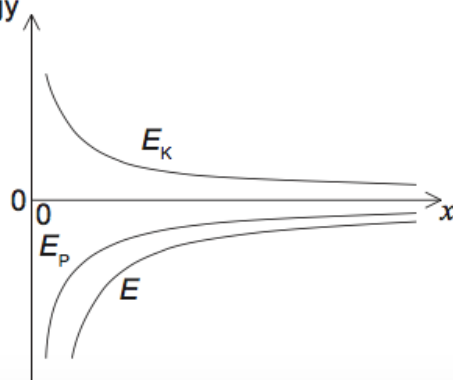
C.

energy



D.

energy



Markscheme

B

28. Which of the following experiments provides evidence for the existence of [1 mark] matter waves?
- A. Scattering of alpha particles
 - B. Electron diffraction
 - C. Gamma decay
 - D. Photoelectric effect

Markscheme

B